Review
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propaganda they invented in an effort to stave off governmental controls. Thus any of these reasons could explain the growth in corporate contributions Nelson ascribes to a greater corporate propensity to give.

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Time Series Analysis, Forecasting, and Control. By GEORGE E. P. BOX and GWILYM M. JENKINS. San Francisco: Holden-Day, Inc., 1970. Pp. xix+553. $24.00.

Practical methods of forecasting large numbers of detailed individual time series, particularly at the level of the individual firm, have long been the province of the fellow with the clear plastic ruler, the clever "eyeball," and the ingenious rule of thumb. In the last thirty years ever-more-sophisticated and ever-larger econometric models have been constructed for the analysis and prediction of major economic aggregates, first at the national level and more recently at the industry level. It is possible to devise "links" between sophisticated econometric models and, for example, the sales of individual items, often numbering into the thousands, produced by a particular company. Such link models are costly to construct, require considerable analytic skill, and frequently perform less well than more naïve forecasting schemes which take into account more adequately serial dependencies present in the individual series to be forecast, and do so without an elaborate causal analysis. The procedures for the analysis and forecasting of time series described by Box and Jenkins in this volume are almost entirely of this "naïve" type. In econometrics, forecasting this period's value or forecasting that the change from this period to the next will be the same as the change from this to the next is called a "naïve model." Pursuing this terminological logic, we might, in somewhat contradictory terms, call the methods suggested by Box and Jenkins "super-sophisticated naïve models."

The methods discussed by Box and Jenkins are nearly all based on fitting a mixed moving-average, autoregressive scheme to each individual time series to be forecast or to some suitable trasnformation thereof. Only in the final third of their book do Box and Jenkins discuss methods applicable to two or more time series. The latter methods are clearly less well developed than the former and occupy a less useful and distinctly secondary role in the authors' minds.

Consider a single, observed time series (x_t), or $(\ldots x_{-1}, x_0, x_1, \ldots)$ and another, unobserved series (ε_t) or $(\ldots \varepsilon_{-1}, \varepsilon_0, \varepsilon_1, \ldots)$. The series (x_t) is said to be generated by a first-order moving-average scheme if, for example,

$$x_t = \epsilon_t + b_1 \epsilon_{t-1}, \tag{1}$$

where the individual ε's are uncorrelated for different values of t and have a common variance. Such a series is usually called "white noise." Similarly, the series is said to be generated by a first-order autoregressive scheme, if

$$x_t - a_1 x_{t-1} = \epsilon_t, \tag{2}$$

where the series (ε_t) has the same properties assumed above. In general, a time series (x_t) is said to be generated by a mixed autoregressive, moving-average scheme (ARMA, for short), if

$$\sum_{j=-m}^{n} a_j x_{t-j} = \sum_{j=-p}^{q} b_j \epsilon_{t-j}, \tag{3}$$

where it is usual to normalize $a_0 = 1 = b_0$. Typically, both the autoregressive and the moving average in (3) are one-sided, so that $m = 0 = p$. The transformations employed involve removing the sample mean or differencing the observed series in order to ensure stationarity of the series to be analyzed. Schemes such as (3) are called "finite parameter" schemes or processes with a "rational spectral density" in the statistical literature of time series analysis.

The central problems involved in forecasting from ARMA models are: first, the choice of transformation and of the orders of the autoregression and moving-average parts of (3), called "model identification" by Box and Jenkins; second, estimation of the parameters involved, or fitting the model; third, analysis of the goodness of fit and the question of whether the correct model has been chosen, or diagnostic checking; and, finally, the design of computationally efficient forecasting procedures based on the estimated model. Box and Jenkins discuss all of these problems in detail for single time series models and in a more tentative and less-detailed way for the bivariate and multivariate cases.

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Part I, chapters 2–5, provides background material on stationary time series and certain nonstationary time series that can be transformed by differencing so as to be stationary. The autocorrelation, partial autocorrelation, and spectral density functions of a stationary time series are introduced. Autoregressive, moving-average, and mixed models and their autocorrelation, partial autocorrelation, and spectral density function are discussed. Autoregressive relationships of the form

$$x_t - x_{t-1} = \epsilon_t \tag{4}$$

are nonstationary since they are characterized by moving averages of white noise which do not give ever-decreasing weight to the more distant past. Linear nonstationary models of this type are called "integrated moving-average processes" by the authors who consider still more general forms of such processes. Because the nonstationarity involved, however, may be eliminated, even in the most complex cases by successive differencing, the theory of model fitting and forecasting may easily be adapted to handle such series. The section closes with a discussion of forecasting time series generated by linear models of the sort previously discussed on the assumption that the parameters of the process are known exactly. For linear stationary models, the minimum mean-square error (M.S.E.) forecast for period $t + \tau$ made at time t is just the conditional expectation of the variable being forecast. This means that the M.S.E. forecasts may always be calculated by replacing future values of observed values by their conditional expectations or M.S.E. forecasts and of future white noise values by zero, which are their conditional expectations. Since it turns out that the white noise appearing in any linear stationary model is nothing but the one-step prediction error of an M.S.E. forecast for t made at time $t - 1$, past values of the ε's are replaced by these errors. Of course, past values of the observed variable are used directly. This device makes it possible to compute forecasts for nonstationary series involving an integrated moving average in like manner, although the authors never explain in what sense such forecasts are optimal. (An early example of the same approach, along with a discussion of optimality, may be found in Couts, Grether, and Nerlove.)[1]

Part II, chapters 6–9, covers the problems of model formulation, or, in Box-Jenkins terminology, identification, estimation, and diagnostic evaluation.

The principal tools suggested for the determination of the orders of the moving-average and autoregressive components of the model to be estimated and of the appropriate differencing transformation to be employed are the estimated autocorrelation and partial autocorrelation functions, particularly the values of these functions for long lags. In view of the well-known statistical deficiencies of the autocorrelation estimates for long lags,[2] it is surprising that Box and Jenkins recommend reliance on these estimates for model identification. Presumably, if the orders of the moving-average and autoregressive components were small relative to the length of the time series analyzed, these difficulties would not be serious; however, economic time series are generally short relative to the complexity of the processes needed to characterize them, so that this aspect of Box and Jenkins' treatment cannot be regarded as satisfactory in an economic context. It is possible to use spectral techniques to obtain a more adequate solution to the problem of model identification.[3] In their discussion of multiple time series and the estimation and identification of transfer function models (Part III, appendix 11.1), Box and Jenkins do suggest use of the cross-spectral density function in passing, but they do not accord such techniques much importance or explore the implications of the spectral density function for model identification in the univariate case.

Methods for estimation of ARMA models, or "finite-parameter" schemes, or models of processes giving rise to time series with rational spectral densities are discussed in detail by Hannan.[4] Spectral methods, which are rela-

[1] D. Couts, D. Grether, and M. Nerlove, "Forecasting Non-stationary Economic Time Series," *Management Science* 13 (September 1966): 1–21.

[2] See, e.g., G. M. Jenkins and D. G. Watts, *Spectral Analysis and Its Applications* (San Francisco: Holden-Day, Inc., 1968), pp. 174–83.

[3] M. Nerlove, "On the Structure of Serial Dependence in Some U.S. Price Series" (Report 7102 of the Center for Mathematical Studies in Business and Economics, University of Chicago, January 1971).

[4] E. J. Hannan, *Multiple Time Series* (New York: John Wiley & Sons, 1970), pp. 325–414.

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tively nonparametric, offer great attractions when the number of observations is large, but as Hannan writes, "if the amount of data available is small there is a great reward to be obtained from a successful investment in a 'finite parameter' model." Inference for such models is largely of a maximum-likelihood character based on normality assumptions but need not, as Hannan points out, be restricted to the "time domain." The Box-Jenkins procedure for the estimation of ARMA models is an approximate maximum-likelihood procedure which makes direct use of the original observations and is said, therefore, to be in the time domain. Their procedure is approximate in two senses: First, the likelihood function itself is approximated by ignoring all but the quadratic term depending on the observations, an approximation valid when the number of observations is moderate to large but which can be seriously misleading when the number of observations is small. Second, the sum of squares appearing in the likelihood function is itself approximated by making use of certain approximations to the expectations of initial unobserved values of the process itself and of its innovations, conditional on the observed sample values. These approximations not only require for their validity that the sample size be fairly large but also that the orders of the moving average and autoregressions be rather low and the autocorrelation function strongly damped. None of these things can be counted on as a matter of course in the study of economic time series. An appendix discusses extensions of the method, but these do not appear computationally practical.

Other methods for estimating the parameters of mixed moving-average, autoregressive processes are discussed by Hannan.[5] These include Durbin and Walker's early suggestions, as well as Hannan's more general procedure using spectral methods. Use of spectral methods simplifies the maximum-likelihood computations, since the use of the Fourier transforms of the original data approximately diagonalizes the variance-covariance matrix of the transformed data. Unfortunately, Hannan's procedures may well be subject to many of the same difficulties in small samples as are the Box-Jenkins procedures. Nonetheless, development of a practical method based on Hannan's suggestions ought to be worthwhile if only because of the potential of spectral methods in model identification.

Once the parameters of a particular model have been estimated, it can be used for forecasting as well as a variety of other purposes. The success of forecasts, or lack of success, is, of course, one of the best checks on a model. However, it appears frequently to be the case, especially for time series containing a strong seasonal component, that the forecasts generated by a model may be quite adequate without its other implications being so at all.[6] Hence, there is a need for additional checking based on the analysis of residuals, alternative model formulations, and the like. Box and Jenkins discuss such problems in a relatively short chapter (pp. 285–99). To the extent that their suggested procedures rest on estimates of residual autocorrelations of residuals at long lags, they may be regarded as deficient in an economic context. Moreover, the procedures they discuss will not generally uncover inadequacies in the transformations of the original data designed to achieve stationarity.

Part III, chapters 10 and 11, discusses transfer function models and all of the problems of identifying, estimating, and checking such models. The simplest form of a transfer function model relating two series (y_t) and (x_t) is the so-called linear filter, e.g.,

$$y_t = a_0x_t + a_1x_{t-1} + \ldots + a_mx_{t-m}\,, \quad (5)$$

more familiar to economists as a "distributed lag." As is well known from the econometric literature, serious problems arise when there is a disturbance, or noise, in the relationship connecting y and x and that disturbance is serially correlated. Box and Jenkins devote an entire chapter to the identification, estimation, and checking of transfer function models with noise. They do not, however, offer many insights additional to those available in the econometric literature.[7]

Throughout most of their book, Box and Jenkins are concerned with the modeling of discrete time series, either individually or severally, and the applications of such models

[5] Ibid., pp. 372–98.

[6] See D. M. Grether and M. Nerlove, "Some Properties of 'Optimal' Seasonal Adjustment," *Econometrica* 38 (September 1970): 682–703, esp. pp. 691–93.

[7] For a survey, see P. J. Dhrymes, *Distributed Lags* (San Francisco: Holden-Day, Inc., 1971).

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to the forecasting of time series by their own part or by means of "leading indicators." In the final section of their book, Part IV, chapters 12–13, the authors discuss wider fields of application in dynamic optimization and control of discrete systems. Perhaps Phillips's early paper[8] first attracted significant attention among economists to this important area, but it is of increasing interest among both economists and operations researchers. The two chapters of Part IV may be especially recommended as a clear and accessible introduction.

On net balance, however, I would not call this a very successful book: it is uneven in level; it tries to give elementary expositions of complex problems which wind up as almost unintelligible; it is often frustrating and, at times, brilliant. But the book is unquestionably an important contribution, especially to the theory of estimating mixed moving-average, autoregressive models.

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Economic Theory and the Antitrust Dilemma. By PETER ASCH. New York: John Wiley & Sons, 1970. Pp. x+414. $9.95.

Peter Asch writes in the preface to this excellent text that "The purpose of this book . . . is not merely to 'cover' antitrust materials, but to indicate something about the potential and actual contribution of economic analysis to policy formulation." He succeeds in achieving both goals. Indeed, he is so successful that this book seems a likely candidate to become the standard text for undergraduate industrial organization courses with an antitrust emphasis.

Asch divides his book into three parts. Part I, titled "The Theoretical Foundations of Antitrust," presents a succinct (130 pages) but informative discussion of the economic theory necessary to analyze antitrust problems. Besides presenting a traditional formulation of the relevant parts of the theory of the firm, Part I presents extensive discussions about both (1) the efficiency implications deductible from welfare economics and (2) the qualifications that one ought to consider before using the implications of either welfare economics or the theory of the firm as the basis for analyzing specific public policy problems.

My only criticism of Part I stems from the fact that the discussion of economic theory is not completely self-contained. For example, Asch uses the concept of elasticity without defining it; also he merely asserts that profits are maximized when a firm produces that output where marginal revenue equals marginal cost. Therefore, readers with poor backgrounds in micro-theory will need to make occasional references to an elementary price-theory text.

Part II, titled "Empirical Problems and Evidence," critically evaluates the ability of economic theory "to predict or explain actual events." This section contains excellent discussions about what phenomena those interested in antitrust problems ought to measure and the empirical problems they must face when they try to measure them.

Part III, titled "Antitrust Policy," first discusses the key sections of the principal antitrust laws of the United States. Then it summarizes numerous court rulings in order to illustrate the evolving interpretations about what is the content of these laws. This extensive summary is both well written and balanced. Part III concludes with a balanced critique of current and past antitrust policies.

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Business Corporations and the Black Man. By S. PRAKASH SETHI. Scranton, Pa.: International Textbook Co., 1970. Pp. xviii+184. $6.00 (cloth); $3.25 (paper).

At the time Mr. Sethi's book came across my desk, the contest for mayor of Chicago was under way. The Democratic candidate, Richard J. Daley, was running for his fifth term and had received the endorsement, among others, of Chicago big business. He won the election, of course. The black precincts again voted for Daley and his margin was 417,000 votes over an energetic Republican challenger who had tried to form a coalition of blacks, liberals, and Republicans.

The people seemed to have voted for the hard line. They wanted plate-glass insurance against the Abbie Hoffmans, the Weathermen, the black rioters of the West Side, the Black

[8] A. W. Phillips, "Stabilization Policy in a Closed Economy," *Economic Journal* 64 (1954): 290–323.

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